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A Conversational Model for Indian Sign Language: From Static Gestures to a Conversational Prototype

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***Abstract*—Indian Sign Language (ISL) enables communication between Deaf and hearing-impaired people, but interacting with people not familiar with ISL is difficult. This paper proposes a framework for camera-based recognition of ISL that includes two phases: recognition of static alphabet and digits, followed by isolated word recognition as a stepping stone towards interactive conversation. In the case of word-level recognition, a vocabulary containing 22 classes was chosen from the "Greetings and Days and Time" categories of the INCLUDE dataset [1]. MediaPipe Holistic tool was used for extraction of landmarks for hands and poses from the video sequences. The extracted landmarks were converted to body-centered and normalized landmarks with respect to scale, and velocity and acceleration were added as features representing movement. Then, a combination of CNN, BiGRU, and attention-based temporal pooling networks was used for classification of the obtained sequence. Word-level recognition model has shown 92.59% accuracy on the held-out test set. Also, a prototype of real-time recognition system using the webcam and the same preprocessing pipeline was implemented with the use of pre-trained Tensorflow Lite model. Preliminary webcam experiments with the webcam demonstrated discrepancies between recognition on the dataset videos and webcam input, reflecting some difficulties connected with variations of the signers and recording conditions. Proposed framework paves the way to further work such as continuous sign segmentation, formation of sentences, language normalization, and speech-based conversational interaction.**

***Index Terms*—Indian Sign Language, Sign Language Recognition, MediaPipe Holistic, Landmark-based Recognition, Temporal CNN, BiGRU, Deep Learning, Assistive Technology, Real-Time Recognition**

I. Introduction

Communication with people unfamiliar with Indian Sign Language (ISL) remains a challenge for many Deaf and hard-of-hearing people. ISL serves as an important medium of communication within the Deaf community, allowing people to express ideas, emotions, and information through hand gestures and body movements. However, communication barriers persist because many people are unfamiliar with ISL, and access to trained interpreters remains limited.

Recent developments in computer vision and artificial intelligence have created new opportunities for automated sign language recognition. These technologies can help reduce communication barriers by translating hand gestures into text or speech. Although extensive research has been conducted on American Sign Language (ASL), comparatively fewer studies have focused on ISL [4], [8]. Since ISL has its own vocabulary and linguistic structure, approaches developed for other sign languages cannot always be directly applied to it.

Many sign language recognition systems rely on specialized devices, such as sensor-equipped gloves or depth cameras [12]. Although these approaches can achieve good accuracy, they may increase costs and limit accessibility. Webcam-based systems offer a practical alternative because they use commonly available hardware. Advances in frameworks such as MediaPipe have also made hand and pose landmark extraction more accessible for real-time applications [10].

The initial stage of this work focused on recognizing isolated ISL alphabet and digit gestures using a standard

webcam. This established the basic landmark extraction and classification pipeline. The present work extends this framework from static gesture recognition to isolated word-level recognition, providing an intermediate step toward continuous and conversational ISL understanding.

The extended system uses the INCLUDE dataset [1], focusing on a selected vocabulary from the Greetings and Days and Time categories. Instead of relying directly on image pixels, the system represents video sequences through pose and hand landmarks. Temporal features are also used to notice movement over a few frames one after another. The resulting framework gives a starting point for future work on continuous sign segmentation, sentence formation, language normalization and speech-based interaction. The framework also supports sign segmentation, sentence formation, language normalization and speech-based interaction, in later stages.

II. Related Work

Researchers have explored different methods for sign language recognition and generation, including computer vision, machine learning, deep learning, and natural language processing [4].

The system described in [2] focuses on sign language generation based on Indian Sign Language grammar. The study shows why it is important to think about the grammar of ISL when making sign language systems. This matters for our work because knowing how ISL grammar works will be key when making the current recognition system able to handle sentences.

The sociolinguistic study mentioned in [15] looks at how Indian Sign Language's used in communication among the Deaf community. It gives background for understanding the communication problems that drive our research.

The ISLTranslate dataset mentioned in [13] was made for translating Indian Sign Language. The research shows why it is important to have datasets that are made specifically for ISL. These kinds of resources are helpful for building recognition and translation systems that can eventually work with more complex sign sequences.

The systematic review in [4] looks at ways to recognize sign language, including computer vision, machine learning and deep learning. It also talks about the problems that current systems face giving a summary of past work and pointing out places where more progress is needed.

Earlier studies have helped with sign language recognition, translation and making datasets [2], [4], [13], [15].. Recognizing single signs is only part of creating a system that supports real communication. Our work is about taking ISL gesture recognition and making it work for individual words using landmark-based features and time-based deep learning. This sets the stage for work, on continuous sign recognition, sentence building and conversations.

III. Core Models and Technologies

The system extends by using landmark-based computer vision taking time-based features and using learning to recognize isolated ISL words. The main parts are MediaPipe [10] body-centered landmark normalization time-based movement features, a time convolutional network part, a bidirectional GRU and attention-based time pooling.

A. Body-Centered and Scale-Normalized Representation

It is seen that raw image coordinates change a lot when the signer moves with respect to the camera. The current preprocessing step helps by moving the landmarks so that they sit at the point between the shoulders. After that the landmarks are scaled using the distance between the shoulders as a body-scale reference. Hand landmarks are shown in two ways: one set shows the hand landmarks relative to the body and another set shows the hand landmarks relative to the wrist that belongs to that hand. The model also keeps a record of whether the hand and the right hand are visible, in each frame.

B. Temporal Features

Many ISL signs involve hand movements, so recognizing them requires information from multiple video frames. For each sequence, velocity and acceleration are calculated from consecutive normalized landmark frames. These features provide information about the direction and rate of movement in addition to the spatial configuration of the hands and body [3].

C. Temporal CNN

Temporal convolution layers analyze movement patterns across consecutive frames. These layers help the model identify short term changes in hand position and movement, which are important for distinguishing dynamic signs.

D. Bidirectional GRU

A bidirectional GRU processes the temporal sequence in both forward and backward directions. This allows the model to use information from earlier and later frames when determining the representation of a sign.

E. Attention-Based Temporal Pooling

Not every frame in a sign video carries the same amount of information. Attention-based temporal pooling assigns greater importance to informative portions of the sequence before the final classification layers produce the probability distribution over the target vocabulary [8].

F. Classification and Decision Rule

Static and word-level recognition share the same underlying decision logic but differ in what is classified at each step.

Static letter/digit recognition. For a single frame, the 21 hand landmarks extracted by MediaPipe are normalized relative to the wrist and scaled by the palm width to remove dependence on hand size and position. This normalized feature vector is passed through a fully connected classifier trained with categorical cross-entropy loss.The output layer produces a softmax probability distribution over 35 classes, including 26 alphabet gestures (A–Z) and nine digit gestures (1–9), and the predicted label $\hat{y}$ is obtained as

$$\hat{y} = \arg\max_i P(y_i \mid \mathbf{x}_{\text{frame}}), \tag{1}$$

TABLE I: Comparative Summary of Existing ISL / Sign Language Recognition Approaches

Paper / Study	Dataset Used	Model	Parameters	Evaluation Metrics	Result	Limitations	Ref.
ISL Hand Detection using YOLOv5	ISL Alphabet Dataset, Custom ISL Videos	YOLOv5	Learning rate, batch size, confidence threshold, IoU	Precision, Recall, mAP, FPS	mAP > 90%, ∼25–30 FPS (real-time)	Poor performance in low lighting & occlusion	[5]
Real-time Hand Detection using MediaPipe	Custom ISL Dataset	MediaPipe Hands	21 keypoints, frame rate tuning	Precision, FPS	∼30 FPS, lightweight & fast	Not robust in complex backgrounds	[10]
ISL Detection using Faster R-CNN	Kaggle ISL Hand Gesture Dataset	Faster R-CNN	Anchor size, learning rate	Precision, Recall, mAP	High detection accuracy (∼88–92%)	Computationally expensive	[14]
ISL Alphabet Recognition using CNN	ISL Alphabet Dataset	CNN	Epochs, filters, kernel size, dropout	Accuracy, Precision, Recall, F1-score	Accuracy ∼95%	Works only for static gestures	[6]
Dynamic ISL Recognition using CNN+LSTM	Self-recorded ISL Dataset	CNN + LSTM	Sequence length, hidden units	Accuracy, F1-score	∼92% accuracy	Needs large labeled video data	[3]
ISL Recognition using 3D-CNN	ISL CSL Dataset	3D-CNN	Temporal window size, stride	Accuracy	∼94% accuracy	High computational cost	[9]
ISL Sentence Translation using BiLSTM	ISL Sentence Dataset	CNN + BiLSTM	Hidden units, embedding size	WER, Accuracy	WER ∼18%	Poor long sentence translation	[7]
Transformer-based ISL Translation	Word-level ISL Dataset	Transformer (Seq2Seq + Attention)	Layers, attention heads	BLEU, WER	BLEU ∼30–35	Requires large annotated dataset	[8]
ISL Interpretation System	ISL Video + Text Corpus	CNN + LSTM + NLP + TTS	Speech rate, decoding length	Accuracy, BLEU, Latency, User Satisfaction	Real-time prototype (∼1–2 sec latency)	Limited vocabulary	[11]

where $\mathbf{x}_{\text{frame}}$ is the normalized landmark feature vector for that frame. To reduce flicker from single noisy frames during real-time use, predictions are smoothed by majority voting over a short sliding window of consecutive frames before being accepted as the displayed letter.

Word-level recognition. As described in Section V-D, a 40-frame sequence is passed through the temporal CNN, Bi-GRU, and attention-pooling stack to produce a single softmax distribution over the 22 word classes. The predicted word $\hat{w}$ is

$$\hat{w} = \arg\max_{j} P(w_j \mid \mathbf{X}_{1:40}), \tag{2}$$

where $\mathbf{X}_{1:40}$ is the full 40-frame, 1,158-feature input sequence. A prediction is only accepted if its confidence exceeds a fixed threshold (empirically set to 0.5 in the current prototype); predictions below this threshold are treated as *no sign detected* rather than forced into the nearest class, which reduces false positives during idle or transitional webcam frames.

Algorithm 1 summarizes the shared decision procedure used at inference time for both the static and word-level classifiers.

Algorithm 1 Inference-time classification decision rule

1: **Input:** feature representation $\mathbf{x}$ (single-frame vector for static gestures, or 40-frame sequence for words)
2: **Input:** trained classifier f_θ, confidence threshold τ
3: $P \leftarrow f_\theta(\mathbf{x})$ {softmax probability vector}
4: $i^* \leftarrow \arg\max_i P_i$
5: **if** $P_{i^*} \geq \tau$ **then**
6: **output** class i^* with confidence P_{i^*}
7: **else**
8: **output** "no sign detected"
9: **end if**

G. TensorFlow/Keras and TFLite

TensorFlow and Keras built and trained the model. We converted it to TensorFlow Lite for lightweight inference during webcam testing, which let us evaluate the same recognition model outside the training environment. This also opens a path to Android deployment later.

IV. Dataset

The word level extension is based on INCLUDE dataset [1] which is a publicly available dataset for isolated Indian Sign Language recognition. It has 4,287 videos that show 263 signs across 15 categories. Since the videos show signs, the dataset is good for building a word-level recognition model before working on continuous sign language translation.

For this prototype, two categories were selected: Greetings and Days and Time. At first 24 classes were looked at. 22

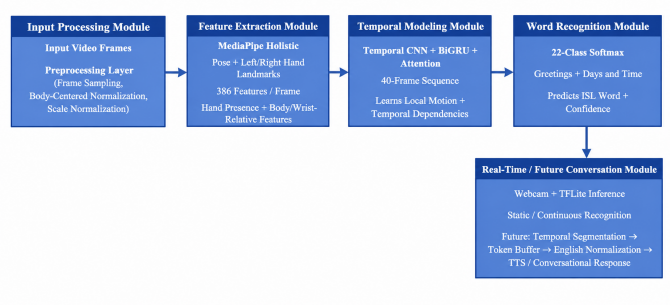


Fig. 1: ISL recognition architecture and conversational pipeline design in the future.

Were kept after checking the samples in the training, validation and test splits. The selected vocabulary includes common expressions like hello, thank you, good morning, good night, today, tomorrow, Monday and Sunday.

After preprocessing and class filtering, the experiment used 372 sequences for training, 38 for validation, and 83 for testing. This focused subset allowed the complete pipeline, from video input to model inference, to be developed and evaluated. Future work will add words and include more examples and different ways of signing and recording conditions.

V. System Architecture

A. Data Collection

For recognition at the word level, video samples were taken from the INCLUDE dataset [1]. Selected video samples were split into training, validation and test datasets.

B. Preprocessing

The input preparation includes video pre-processing that provides a consistent input for model training. Landmarks for pose and hands are obtained from single frames. These features are normalized to eliminate variance in signers' position. The result is put in a form of a fixed-length sequence of landmarks for recognition.

C. Feature Extraction

Pose and hand landmarks are found with MediaPipe Holistic library, while OpenCV is used for reading videos and handling frames [10]. Instead of processing images through the classifier, only landmarks that describe pose and location of hands are used as an input to obtain compact representation for further recognition.

D. Model Development

Sequence of 40 frames with 1,158 features is fed to the word-recognition model. Two temporal convolutional layers capture movement in frames that lie next to one another. This is followed by bidirectional GRU to capture temporal dependencies in both directions. Attention-based pooling finds most informative frames. Fully connected layers and softmax provide classification for 22 word classes.

The model can be summarized as:

$$\begin{aligned} 40 \times 1{,}158 \text{ sequence} \rightarrow & \text{Conv1D} \rightarrow \text{BatchNorm} \rightarrow \\ & \text{Conv1D} \rightarrow \text{BatchNorm} \rightarrow \\ & \text{BiGRU} \rightarrow \text{Attention Pooling} \rightarrow \\ & \text{Dense} \rightarrow \text{Softmax} \end{aligned} \tag{3}$$

Output of the word-recognition model is a probability distribution over 22 word classes.

E. Training and Validation

The model was trained with the help of the training set created above, while the validation set helped to monitor its performance during training. The separate test set was employed in evaluation of recognition accuracy in unseen samples.

F. Gesture Recognition

The model can operate in the static-window and continuous webcam modes. For the first mode, a recording from the webcam is taken into a sequence of 40 frames which are then classified once. In the second one, a sliding window containing 40 frames is updated by frames incoming from a webcam during runtime. Temporal consistency rules can be used to smooth down variations in the label display.

G. Text and Speech Generation

The recognized word is displayed in the text form. Speech generation and conversational reply are the future extensions of the proposed pipeline [11].

H. Deployment

The word-level model was converted into TensorFlow Lite for webcam inference. Further development can lead to its extension in order to provide functionality on the mobile platform as well as vocabularies expansion, sentence formation, language normalization, speech generation, and conversation [7].

VI. Results and Discussion

A. Initial Static Gesture Recognition

The initial stage focused on static ISL gesture recognition using a 35-class classifier covering alphabets A–Z and digits 1–9. Under controlled lighting conditions, the model achieved approximately 92% validation accuracy for alphabet recognition and 94% for digit recognition. Under low-light conditions and cluttered backgrounds, the corresponding accuracies decreased to approximately 81% and 84%, respectively. This initial stage established the recognition pipeline for extending the system from static gestures to isolated ISL words.

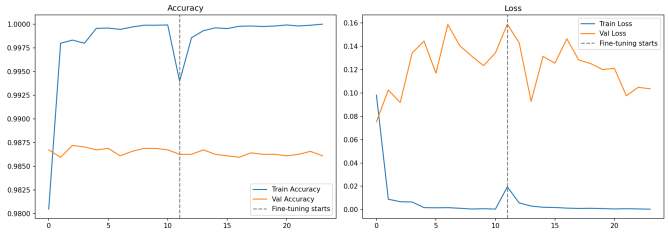


Fig. 2: Training and validation accuracy and loss for the 35-class static ISL alphabet and digit recognition model

TABLE II: Word-Level Predictions on Selected INCLUDE Video Samples

Expected Sign	Predicted Sign	Confidence
Hello	Hello	87.31%
Good night	Good night	45.27%
Good morning	Good morning	45.77%
Tomorrow	Tomorrow	69.90%

B. Word-Level ISL Recognition

The second stage scaled up the system to isolated word recognition with 22 classes drawn from the Greetings and Days and Time categories of the INCLUDE dataset [1]. The model leverages spatial landmarks features along with temporal information to discriminate signs in different video sequences. Held-out test evaluation yields accuracy of 92.59

The exported model in TensorFlow Lite was also tested on individual INCLUDE video samples. All four selected examples yielded the expected labels, as shown in Table II. These checks show that inference on the chosen samples works.

There are different levels of confidence that can be observed among the four predictions: upon matching the labels with their expected signs, we see that both “good night” and “good morning” are operating at low confidence levels, which indicates that one can have a correct predicted label and have low model confidence.

C. Comparison with Existing Approaches

Different techniques used for sign language recognition are diverse from one another according to the input designs employed for recognition as well as architectures used in modeling of the systems, vocabularies employed and methods used to conduct tests. As it can be understood from the descriptions provided by different scholars, static classifiers detect each of the different forms used by a hand, while temporal approaches rely on the information obtained from many different frames [3], [4], [8]. The work represented here is a continuation of the work on static ISL recognition methods that was taken already before towards the recognition of ISL words based on similar features and processes. Because the data used in this research as well as the later settings of this research do not match the ones used here, therefore, it means it would not be possible to perform a fair comparison of the results obtained with those produced in contexts of any previous research.

D. Real-Time Webcam Evaluation

The exported model was incorporated into an inference pipeline using a webcam, which offers both static-window and continuous modes of operation. In static-window mode, a sequence is created out of a short recording and classified only once. On the other hand, continuous mode uses a rolling sequence for multiple classifications while the camera is still on.

The early tests done with webcams indicate that the outcome of dataset-based video predictions is not always applicable for real-life webcam inputs. In some instances, the model was prone to certain categories regardless of the sign used. The variability in the background context, like the signer position, camera distance from the signer, hand visibility, lighting, and signing style could be affecting this behaviour. Although the temporal smoothing helps to minimize deviations from predictions, it does not ensure the absence of errors due to the availability of input videos.

E. Future Work

The existing word-level recognizer utilizes 22 isolated classes of ISL and works as the recognition component of a larger conversation system. The upcoming enhancements will involve expanding the vocabulary, enhancing the recognition of the different signers and environmental factors, and improving the continuous sign segmentation. In addition, the upcoming features will combine the recognized sequences of words with language normalization, sentence structure, speech generation, and conversation response capabilities.

VII. Conclusion

In this research work, the authors have proposed a vision-based framework for Indian Sign Language (ISL) recognition that comprises two phases. The first stage of the researchers has established an efficient static gesture recognition system for numbers and letters based on landmark manipulation. The second stage of the researchers has further developed the system to conduct word recognition using data from the INCLUDE dataset [1].

The existing word prototype can identify 22 ISL classes by utilizing body-centric and size-normalized pose and hand features along with the time sequencing of movements. The word recognition system has a reported accuracy of 92.59% on the test data. Further testing using videos from the dataset has confirmed that the pulled-out TensorFlow Lite model is able to identify the distinct ISL words correctly.

Live webcam tests shed light on a serious issue: that performance with regard to videos of the dataset is not necessarily applicable to webcam input. Signer position, body size, lighting, camera type, hand visibility, and signing style can affect recognition performance, thus addressing these differences is important to achieve good performance in real-life conditions [4].

Further research will involve increasing the vocabulary, bettering continuous sign segmentation, and buffering and normalizing signs into sequences to form sentences. Research will

also involve speech synthesis and conversation management to make it easier for ISL users to communicate with those who do not know sign language. The aim is to develop a system that goes beyond recognizing individual signs to full communication [11], [13].

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